Lathi 20题练习册 第10批

对应 Boulet 前四章

原书英文题干、公式与必要图示 · 无答案版

记号：D = d/dt；差分式中 E 为前移算子，E{y[n]} = y[n+1]；j^2 = -1
u 为单位阶跃，δ 为单位冲激；LTIC / LTID 分别指连续 / 离散时间 LTI 系统

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来源：B. P. Lathi / R. A. Green, Linear Systems and Signals, 3rd ed. (2018)
中文主题仅用于课程对应；题目要求以原书英文为准。

周期三次波的功率、均方根值及幅度缩放

Lathi 1.1-4 | 原书第 136、137 页

Boulet 第1章：Finite-Energy and Finite-Power Signals，p.21；Periodic Signals，p.8

1.1-4 Find the power of the periodic signal $x(t)$ shown in Fig. P1.1-4. Find also the powers and the rms values of:

(a) $-x(t)$

(b) $2x(t)$

(c) $cx(t)$

Comment.

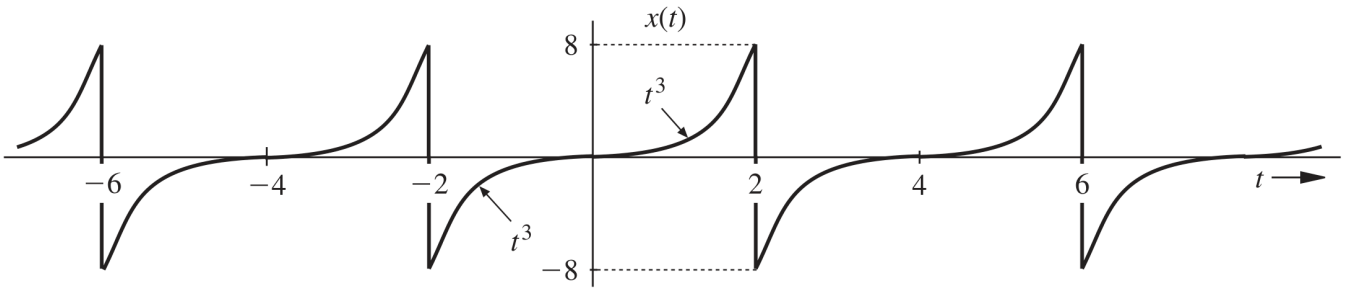


Figure P1.1-4

演算区

信号反转平移与周期复制

Lathi 1.3-2 | 原书第 140 页

Boulet 第1章：Transformations of the Time Variable，p.4；Periodic Signals，p.8

1.3-2 Define signal $y(t) = \sum_{k=-\infty}^{\infty} x(0.5t - 10k)$, where

$$x(t) = \begin{cases} e^{-2t} & t \geq 1 \\ 0 & t < 1 \end{cases}$$

(a) Determine the constant a such that the signal $x(-2t + a)$ is borderline anticausal.

(b) Is the signal $y(t)$ periodic? If so, determine the period T_y. If not, explain why $y(t)$ is not periodic.

演算区

冲激时间缩放后的全轴积分

Lathi 1.4-7 | 原书第 142 页

Boulet 第1章：Generalized Functions, p.26

1.4-7 For real and positive constant a, evaluate the following integral:

$$\int_{-\infty}^{\infty} \delta(at)\, dt$$

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斜坡乘法系统的基本性质

Lathi 1.7-7 | 原书第 145 页

Boulet 第1章：System Models and Basic Properties， p.34

展开所引用1.7-4的要求：分别判断系统是否(a)BIBO稳定、(b)因果、(c)可逆、(d)线性、(e)无记忆、(f)时不变；每项回答是或否，并给出数学论证。

1.7-7 Repeat Prob. 1.7-4 for a system that multiplies a given input by a ramp function, $r(t) = tu(t)$. That is, $y(t) = x(t)r(t)$.

演算区

复频率与离散指数底数的相互表示

Lathi 3.3-8 | 原书第 317 页

Boulet 第1章：Exponential Signals, p.9；Periodic Complex Exponential and Sinusoidal Signals, p.17

3.3-8 Express the following signals, which are in the form $e^{\lambda n}$, in the form γ^n:

(a) $e^{-(1+j\pi)n}$

(b) $e^{-(1-j\pi)n}$

(c) $e^{(1+j\pi)n}$

(d) $e^{(1-j\pi)n}$

(e) $e^{-[1+j(\pi/3)]n}$

(f) $e^{[1-j(\pi/3)]n}$

演算区

周期矩形波经有限斜坡冲激响应的卷积

Lathi 2.4-38 | 原书第 231 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67

所需原图P2.4-38的周期输入x(t)及冲激响应h(t)均完整附上。

2.4-38 The periodic signal $x(t)$ shown in Fig. P2.4-38 is input to a system with impulse response function $h(t) = t[u(t) - u(t - 1.5)]$, also shown in Fig. P2.4-38. Use convolution to determine the output $y(t)$ of this system. Plot $y(t)$ over $(-3 \leq t \leq 3)$.

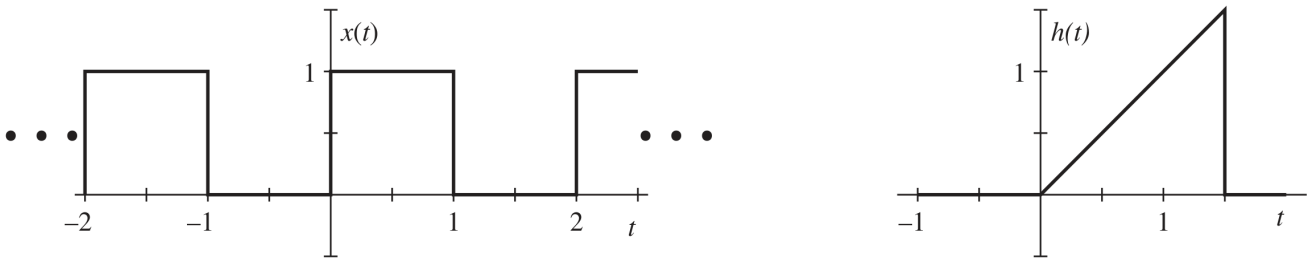


Figure P2.4-38

演算区

有限斜坡输入的卷积输出、稳定性与因果性

Lathi 2.5-8 | 原书第 233 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67；Properties of Linear Time-Invariant Systems，p.74

2.5-8 An analog LTIC system with impulse response function $h(t) = u(t+2) - u(t-2)$ is presented with an input $x(t) = t(u(t) - u(t-2))$.

(a) Determine and plot the system output $y(t) = x(t) * h(t)$.

(b) Is this system stable? Is this system causal? Justify your answers.

演算区

有限序列的幅度缩放、时移与卷积索引

Lathi 3.8-26 | 原书第 324 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54；Properties of Linear Time-Invariant Systems，p.74

保留两组序列的n=0箭头；答案须标明最左侧非零样本的时间索引。

3.8-26 Let $\downarrow$ identify the $n = 0$ and consider DT signals $x[n]$ and $h[n]$ whose nonzero values are given as $x[n] = [1, 2, 3, \overset{\downarrow}{4}, 5]$ and $h[n] = [-2, -1, 1, \overset{\downarrow}{2}]$. Use DT convolution (any method) to determine $y[n] = (2x[n - 30]) * \left(-\frac{3}{2}h[n - 10]\right)$. Express your result in vector notation, making sure to indicate the time index of the leftmost (nonzero) element.

演算区

用滑动序列法证明阶跃与矩形序列卷积

Lathi 3.8-27 | 原书第 324 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54

原题列出的等式是要求证明的目标，并非另附解答。

3.8-27 Using the sliding-tape algorithm, show that

(a) $u[n] * u[n] = (n+1)u[n]$

(b) $(u[n] - u[n-m]) * u[n] = (n+1)u[n] - (n-m+1)u[n-m]$

演算区

由有限输入输出序列反求冲激响应

Lathi 3.8-31 | 原书第 324 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54

保留输入、输出两组序列的n=0箭头。

3.8-31 Letting $\downarrow$ identify $n = 0$, define the nonzero values of signal $x[n]$ as $[1, 2, \underset{\downarrow}{2}]$. Similarly, define the non-zero values of signal $y[n]$ as $[3, 4, 6, 6, 11, \underset{\downarrow}{2}, -2]$. Using the sliding-tape algorithm as the basis for your work, determine the signal $h[n]$ so that $y[n] = x[n] * h[n]$.

演算区

RC电路的阶跃响应与指定时刻初值设计

Lathi 2.4-34 | 原书第 230 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92；Time Constant and Natural Frequency of a First-Order LTI Differential System，p.116

题干(a)、(b)及图P2.4-34全部附上；R、C按原题保留为符号参数。

2.4-34 Consider the circuit shown in Fig. P2.4-34.

(a) Find the output $y(t)$ given an initial capacitor voltage of $y(0) = 2$ volts and an input $x(t) = u(t)$.

(b) Given an input $x(t) = u(t-1)$, determine the initial capacitor voltage $y(0)$ so that the output $y(t)$ is 0.5 volt at $t = 2$ seconds.

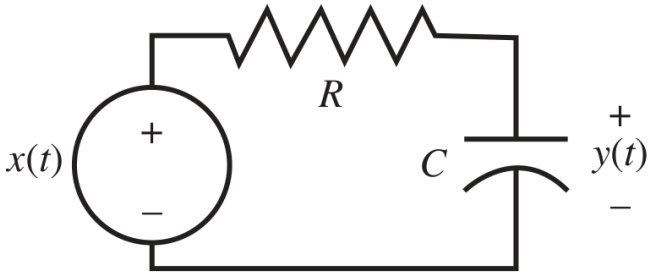


Figure P2.4-34

演算区

电阻与双电感电路的微分模型和零输入响应

Lathi 2.4-36 | 原书第 231 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92；Characteristic Polynomials and Stability，p.112

题干(a)-(c)、电感关系和图P2.4-36完整附上；初始电流均为1 A。

2.4-36 Consider the electric circuit shown in Fig. P2.4-36.

(a) Determine the differential equation that relates the input current $x(t)$ to output current $y(t)$. Recall that

$$v_L(t) = L\frac{di_L(t)}{dt}$$

(b) Find the characteristic equation for this circuit, and express the root(s) of the characteristic equation in terms of L_1, L_2, and R.

(c) Determine the zero-input response given initial inductor currents of one ampere each. That is, find $y_0(t)$ given $i_{L_1}(0) = i_{L_2}(0) = 1$ A.

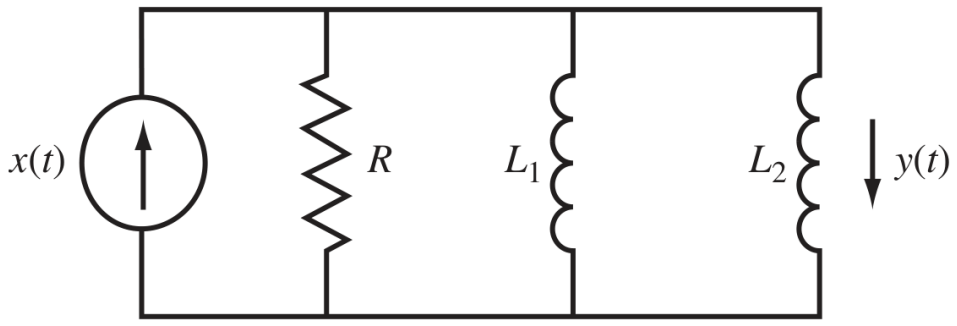


Figure P2.4-36

由二阶微分方程建立差分近似模型

Lathi 3.4-5 | 原书第 317 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92；Causal LTI Systems Described by Difference Equations，p.96

3.4-5 Approximate the following second-order differential equation with a difference equation.

$$\frac{d^2y(t)}{dt^2} + a_1\frac{dy(t)}{dt} + a_0y(t) = x(t)$$

演算区

复系数差分系统的冲激响应与总响应

Lathi 3.8-37 | 原书第 326 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations，p.96；Impulse Response of a Difference LTI System，p.109

记号说明：E是单位超前算子，即E x[n]=x[n+1]（Lathi p.269）；j为虚数单位。

3.8-37 A system is called complex if a real-valued input can produce a complex-valued output. Consider a causal complex system described by a first-order constant coefficient linear difference equation:

$$(jE + 0.5)y[n] = (-5E)x[n]$$

(a) Determine the impulse response function $h[n]$ for this system.

(b) Given input $x[n] = u[n-5]$ and initial condition $y_0[-1] = j$, determine the system's total output $y[n]$ for $n \geq 0$.

演算区

由有限离散冲激响应建立差分方程

Lathi 3.8-38 | 原书第 326 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations，p.96；Impulse Response of a Difference LTI System，p.109

3.8-38 A discrete-time LTI system has impulse response function $h[n] = n(u[n-2] - u[n+2])$.

(a) Carefully sketch the function $h[n]$ over $(-5 \leq n \leq 5)$.

(b) Determine the difference equation representation of this system, using $y[n]$ to designate the output and $x[n]$ to designate the input.

演算区

（复习）周期性、基周期与谐波组成

Lathi 6.1-8 | 原书第 672 页

Boulet 第4章：Linear Combinations of Harmonically Related Complex Exponentials, pp.132-134

复习题：此前见第3批。复习第3批：重练周期性、基周期与谐波组成，注意有理频率比以及三角函数乘方展开后的频率。

6.1-8 State with reasons whether the following signals are periodic or aperiodic. For periodic signals, find the period and state which harmonics are present in the series.

(a) $3\sin t + 2\sin 3t$

(b) $2 + 5\sin 4t + 4\cos 7t$

(c) $2\sin 3t + 7\cos \pi t$

(d) $7\cos \pi t + 5\sin 2\pi t$

(e) $3\cos \sqrt{2}t + 5\cos 2t$

(f) $\sin \frac{5t}{2} + 3\cos \frac{6t}{5} + 3\sin\left(\frac{t}{7} + 30^\circ\right)$

(g) $\sin 3t + \cos \frac{15}{4}t$

(h) $(3\sin 2t + \sin 5t)^2$

(i) $(5\sin 2t)^3$

演算区

（复习）反转、平移、调制与线性组合的级数系数

Lathi 6.3-14 | 原书第 675 页

Boulet 第4章：Properties of Continuous-Time Fourier Series，pp.139-140（Linearity、Time Shifting、Time Reversal、Multiplication of Two Signals）

复习题：此前见第3批。复习第3批：重练时间反转、时移、余弦调制及线性组合。原文先将输入谱记为D1[n]，后文又写X1[n]；两者均指同一输入x1(t)的傅里叶级数系数，作答时统一记号即可。

6.3-14 (a) A 2 rad/s periodic signal $x_1(t)$ has Fourier series spectrum $D_1[n]$. Determine the Fourier series spectrum $X_2[n]$ of $x_2(t) = \frac{1}{3}x_1(-t-5)$ in terms of $X_1[n]$.

(b) A 2 rad/s periodic signal $x_1(t)$ has Fourier series spectrum $D_1[n]$. Determine the Fourier series spectrum $X_2[n]$ of $x_2(t) = \cos(10t)x_1(t)$ in terms of $X_1[n]$.

(c) A 3 rad/s periodic signal $x_1(t)$ has Fourier series spectrum $D_1[n]$. Determine the Fourier series spectrum $X_2[n]$ of $x_2(t) = x_1(-t) - 3x_1(t+2)$ in terms of $X_1[n]$.

演算区

（复习）分段二次周期信号的傅里叶级数与幅度谱

Lathi 6.3-15 | 原书第 675 页

Boulet 第4章：Determination of the Fourier Series Representation of a Continuous-Time Periodic Signal，pp.134-136；Graph of the Fourier Series Coefficients: The Line Spectrum，p.137；Properties of Continuous-Time Fourier Series，pp.139-141

复习题：此前见第3批。复习第3批：重练分段二次周期信号的级数、直流分量与幅度谱。原文引用例6.11仅作方法提示：(c)要求用级数性质而非直接积分求非直流系数；本题信号条件已全部给出，不依赖该例的数据或结果。

6.3-15 A 2-periodic signal $x(t)$ is defined as

$$x(t) = \begin{cases} -t^2 - t + 0.25 & -1 \le t < 0 \\ t^2 - t + 0.25 & 0 \le t < 1 \\ x(t+2) & \forall t \end{cases}$$

(a) Plot $x(t)$ over $-2 \le t \le 2$.
(b) Determine D_0, the dc content of $x(t)$.
(c) Similar to Ex. 6.11, use properties and not integration to determine D_n for $n \neq 0$.
(d) Plot the magnitude spectrum $|D_n|$ over a suitable range of n.
(e) How does the magnitude spectrum $|D_n|$ compare to that of signal $y(t) = \cos(\pi t)$? Note similarities as well as major differences.

演算区

（复习）正弦乘积的级数频谱与理想带通滤波

Lathi 6.4-4 | 原书第 675、676 页

Boulet 第4章：Properties of Continuous-Time Fourier Series，pp.139-140；Graph of the Fourier Series Coefficients: The Line Spectrum，p.137；Steady-State Response of an LTI System to a Periodic Signal，pp.155-156

复习题：此前见第3批。复习第3批：将正弦乘积的级数、线谱与理想带通滤波后的稳态响应连起来。原输入为cos(5t)sin(3t)；下方续接原书下一页的(c)与所需带通图。

6.4-4 (a) Find the exponential Fourier series for a signal $x(t) = \cos 5t \sin 3t$. You can do this without evaluating any integrals.

(b) Sketch the Fourier spectra.

(c) The signal $x(t)$ is applied at the input of an LTIC system with frequency response, as shown in Fig. P6.4-4. Find the output $y(t)$.

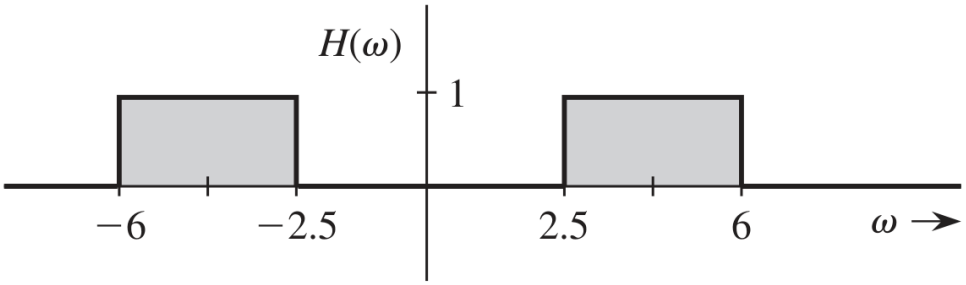


Figure P6.4-4

演算区

（复习）信号的最优投影、误差能量与正交性

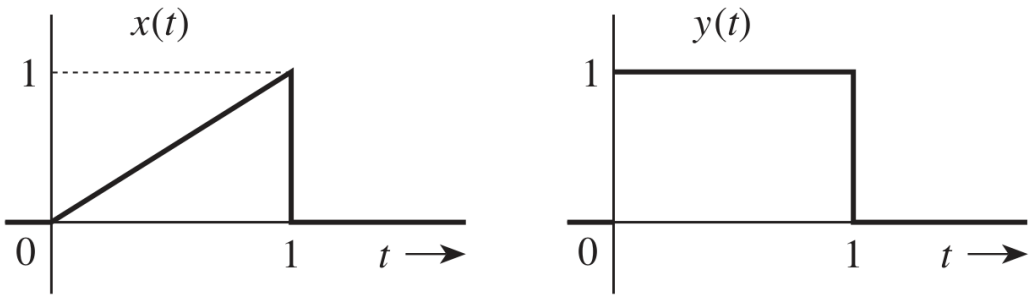
Lathi 6.5-4 | 原书第 677 页

Boulet 第4章：Optimality and Convergence of the Fourier Series，pp.144-145；Parseval Theorem，p.150（正交分量的能量可加性）

复习题：此前见第3批。复习第3批：重练最小误差能量近似、投影系数、误差正交性与能量分解，巩固傅里叶级数最优近似的基础。题中要求验证的能量关系属于原题。

6.5-4 (a) For the signals $x(t)$ and $y(t)$ depicted in Fig. P6.5-4, find the component of the form $y(t)$ contained in $x(t)$. In other words, find the optimum value of c in the approximation $x(t) \approx cy(t)$ so that the error signal energy is minimum.

(b) Find the error signal $e(t)$ and its energy E_e. Show that the error signal is orthogonal to $y(t)$, and that $E_x = c^2 E_y + E_e$. Explain this result in terms of vectors.



(a) (b)

Figure P6.5-4

演算区